benzene (as a solvent) can serve as a protective medium to minimize radiation self-decomposition, whereas water or oxygen enhance it.

Radiation doses of 10^5 Gy can induce decomposition effects of the order of 1%. Samples whose specific activity exceed 40 GBq (1 Ci) per mol for ^{14}C or about 400 GBq (10 Ci) per mol for ^{3}H will receive a dose of this magnitude in a period of a year. Samples may be stored in benzene solution in vacuo or in deep freeze to minimize self-radiation effects and should be re-purified before use if the decomposition products are likely to affect the experiment.

The instability towards irradiation is also a problem for e.g. processes for recycling of nuclear waste by solvent extraction. Then organic molecules are used as extractant and they, together with the organic diluent decompose with time and cleaning stages have to be included. The most famous of these degradation routes is the one relating to the commonly used extraction molecule tri-butyl phosphate (TBP) which form dibutyl phosphoric acid (HDBP) with a yield of approximately $G_{HDBP} = 0.07-0.12$ μmol/J and to a lesser extent bis(1,1,3,3-tetramethylbutyl)phosphinic acid (HMBP). Both these decomposition products are extractants in their own way and disturb the extraction process by increasing the extraction of some unwanted fission and corrosion products.

8.9. EXPERIMENTAL METHODS

Radiation chemistry is characterized by the very fast generation of reactive species followed by extensive competition between recombination reactions and reactions with solutes. A complete description of a radiation chemical process requires information about the final products and the transient species.

The final products can be analyzed with standard chemical methods and much information has been gained through the use of selectively reacting scavengers.

Information about structures and identities of primary ionic species has been obtained from mass spectrometry, photoelectron and vacuum ultraviolet spectroscopy. Electron spin resonance techniques have been used extensively in the study of free radicals. Low temperature, matrixes and "spin traps" have been employed to stabilize the short lived radicals.

Typical irradiation facilities are in the case of gamma radiolysis based on either ^{60}Co or ^{137}Cs. Samples introduced in these, often intense gamma fields (up to tenths of kGy/h), will not become radioactive and the radiolysis products can be studied in normal chemical laboratories. In the case of alpha radiolysis the situation is more complicated. Alpha particles have so short range that external irradiation into a sample vial is almost impossible. Therefore it is common practice to dissolve a alpha radioactive sample into the liquid of interest. The problem is that there will be residual alpha activity present in the sample preventing analysis of degradation product in a non-radioactive laboratory and thus highly sophisticated equipment has to be dedicated and contaminated.

Accelerators producing high energy He beams can also be used because the high energy α-particles will penetrate through the vial walls. Then there will be no residual activity in the sample. Recently a new method based on "in vitro" radiation of solution using ^{211}At has been developed at Chalmers. This method will also produce an, in principle, non radioactive solution with radiolysis products.

Much insight into radiation processes and radiation induced radical reactions has been gained by means of the *pulse radiolysis* method which is based on irradiation of samples with a short pulse of ionizing radiation. The radiation source is generally an electron accelerator or, to a much lesser extent, a positive particle accelerator. Techniques used to follow the transient behaviour of the radiolytically generated short-lived species are optical absorption spectroscopy, esr, conductivity and polarography. The time resolution is generally in the femtosecond to microsecond range.

8.10. DOSE MEASUREMENTS

The amount of radiation energy absorbed in a substance is measured with *dose meters* (or *dosimeters*). These may react via a variety of processes involving (a) the heat evolved in a calorimeter. (b) the number of ions formed in a gas, (c) the chemical changes in a liquid or in a photographic emulsion, and (d) the excitation of atoms in a glass or crystal. The first two ones are *primary meters* in the sense that they can be used to accurately calculate the exposure or dose absorbed from a radiation source. They can be used to calibrate the *secondary meters*.

In 1925 C. D. Ellis and W. A. Woorter, using RaE (^{210}Bi) in *calorimetric measurements*, obtained the first proof that the maximum energy and the average energy of β-radiations were different. A precision of about 1% can be obtained in a calorimeter for an energy production rate of $\sim 10^{-6}$ J/s which corresponds to approximately 0.7 MeV average energy for a sample of 40 MBq (1 mCi). If the average energy of an α- or β-emitting nuclide is known, calorimetric measurement of the energy production rate can be used to calculate the specific activity. This technique is not suitable for γ-sources.

A more sensitive and general instrument for the measurement of ionizing radiation is the *condenser ion chamber.* This is a detector which has a small gas-filled volume between two charged electrodes. When radiation ionizes the gas between the two electrodes, the cations travel to the cathode and the electrons to the anode, thus preventing recombination of the ion pairs. Measurement of the amount of discharge provides a determination of the ionization and consequently of the dose delivered to the instrument. This type of instrument is described in more detail in Ch. 9. The flexibility and accuracy of this dosimeter have led to it being widely employed for the exact measurement of γ-dose rates. The most common version of this type of instrument is the pen dosimeter (Fig. 8.8), which can be made to provide either a direct reading of the absorbed dose or